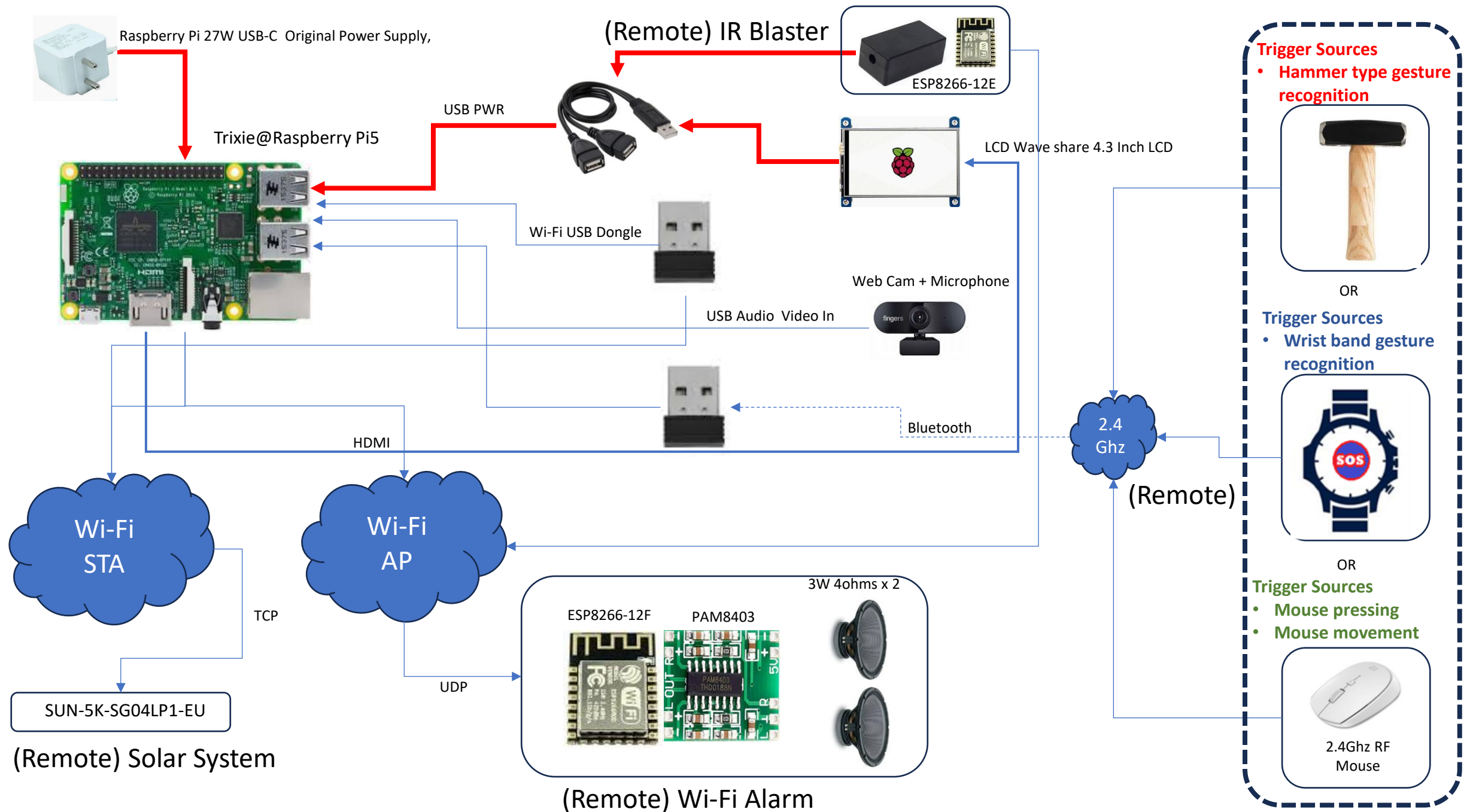


Use Case Wi-Fi Alarm Speaker + [Touch or Gesture or Mouse] Alarm Source + Solar-Grid + IR Blaster





Voice
Activation



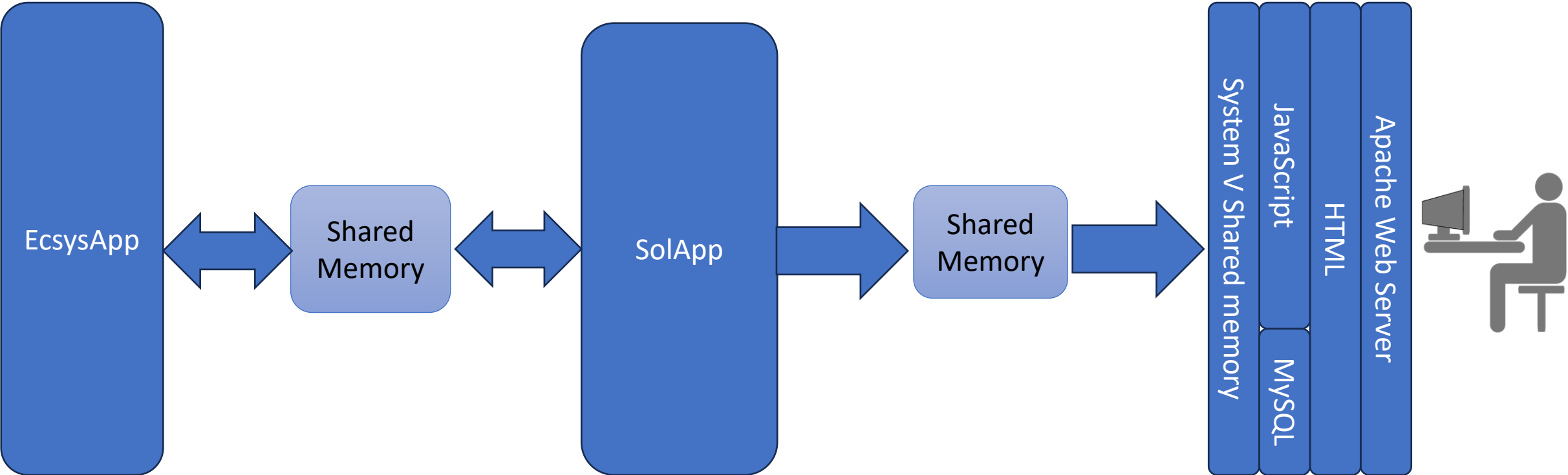
Wrist Band
Activation

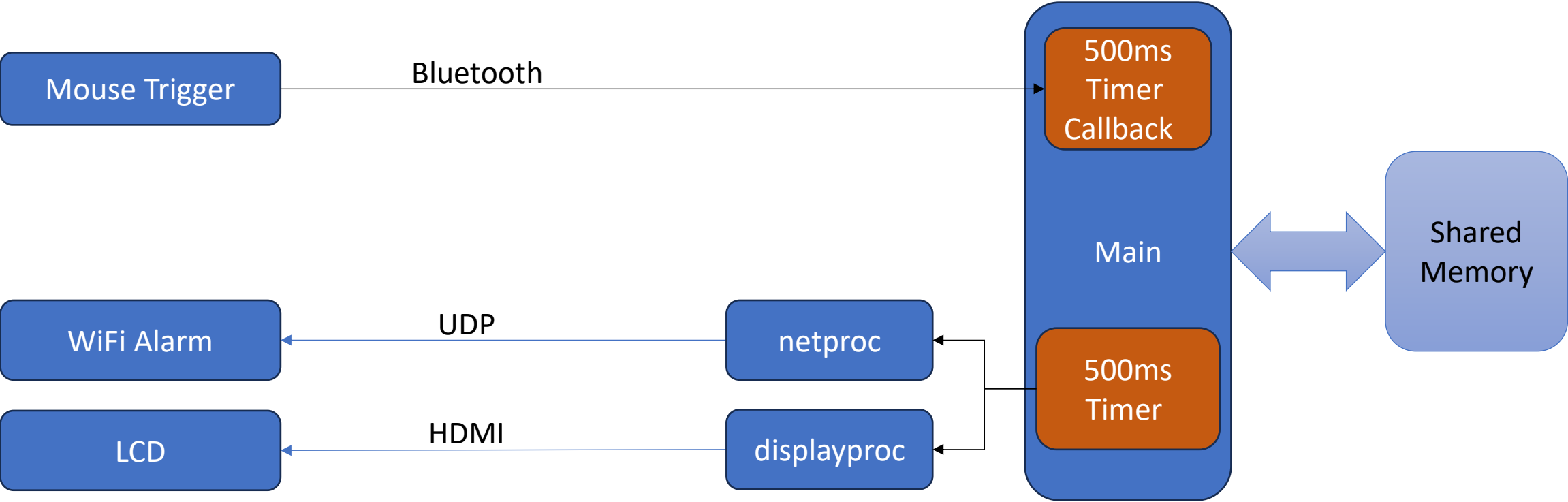


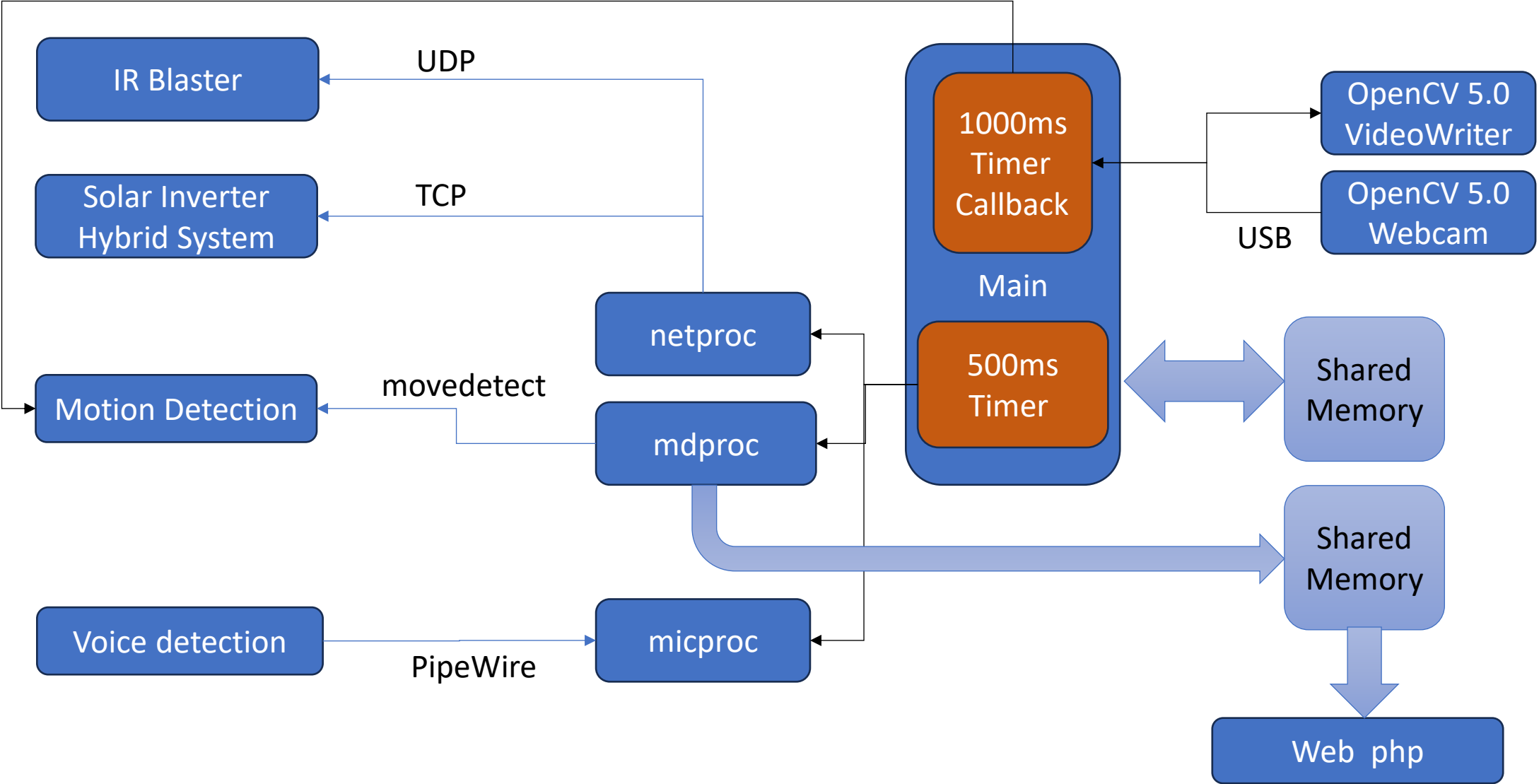
Motion Detection
Activation

ALARM









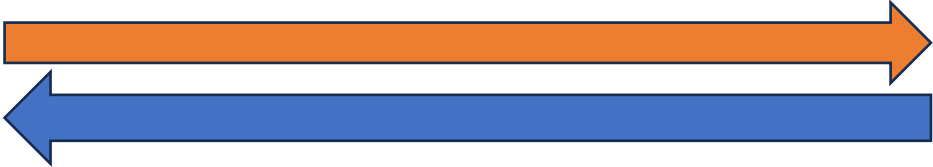
Shared Memory Data

ecsysapp

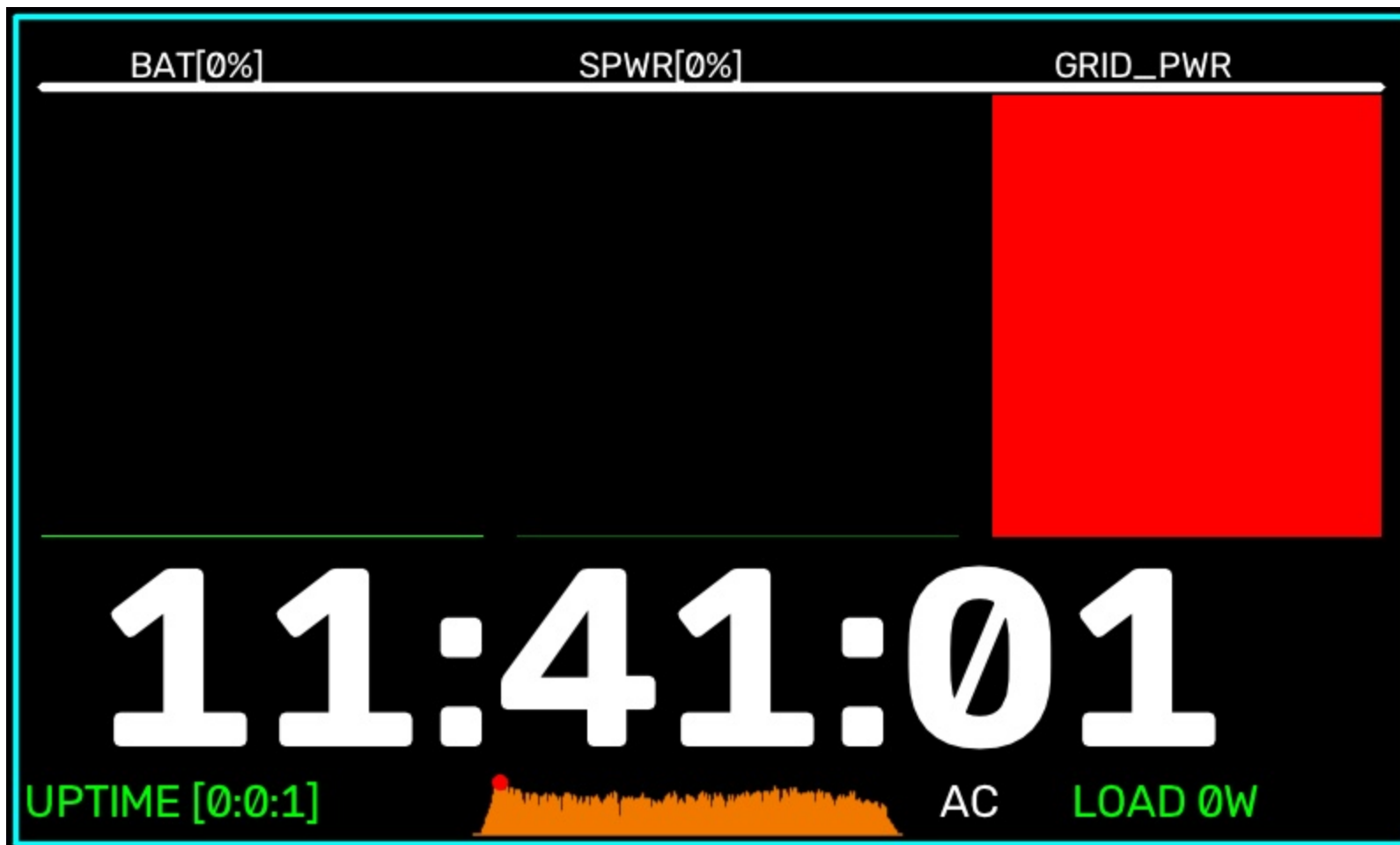
TS
UPTIME
WIFI ALARM
AC
Voice level

solapp

TS
Moton Detection Alarm
Solar Parameters
Voice level
Audio Spectrum
Voice Detection Alarm



LCD screen



System Configurations

Application Parameters

		MOUSE_LEVEL_A[0-5]	0
		MOUSE_LEVEL_B[0-5]	0
		MOUSE_LEVEL_B_HOURS[0,1,2...23]	DEFAULT
		MOUSE_NAME	2.4G Mouse
		MOUSE_INDEX	1
ID	NAME	BEACON_TIMEOUT[1-255]	15
0	2.4G Mouse	DIR_MAX[1-255]	2
1	2.4G Mouse	NIGHT MODE HOURS[0,1,2...23]	20,21,22,23,0,1,2,3,4,5
2	pwr_button	AC ON HOURS[0,1,2...23]	11,13,15,20,22
3	Lenovo FHD Webcam Audio: Lenovo	ALARM KEY	ecsys_key
4	vc4-hdmi-0	ALARM IP	192.168.0.105
5	vc4-hdmi-0 HDMI Jack	VOICE DETECTION HOURS[0,1,2...23]	DEFAULT
6	vc4-hdmi-1	MOTION DETECTION HOURS[0,1,2...23]	DEFAULT
7	vc4-hdmi-1 HDMI Jack	SOLAR IP	192.168.0.100
		IR BLASTER IP	192.168.0.108
		Solar Serial Number	3141237744
		IR BLASTER KEY	ir_key
		ACCESS_PASSWORD	admin

Submit

Select the image area for motion detection

ID22X0Y0W320H480

Mask AddMask Remove

X

Y

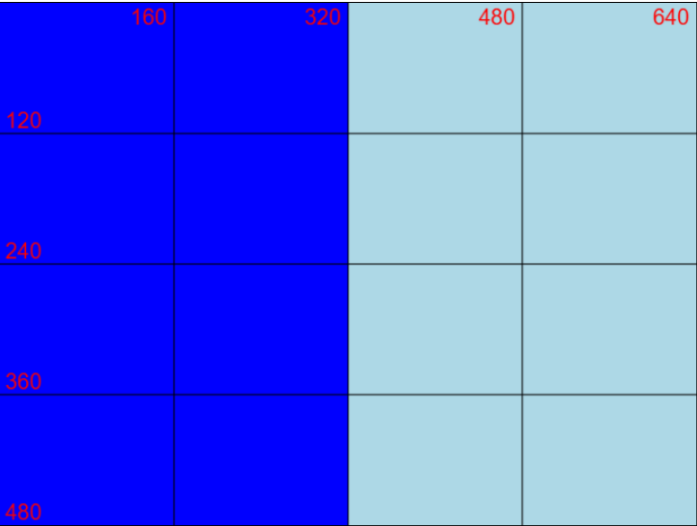
W

H

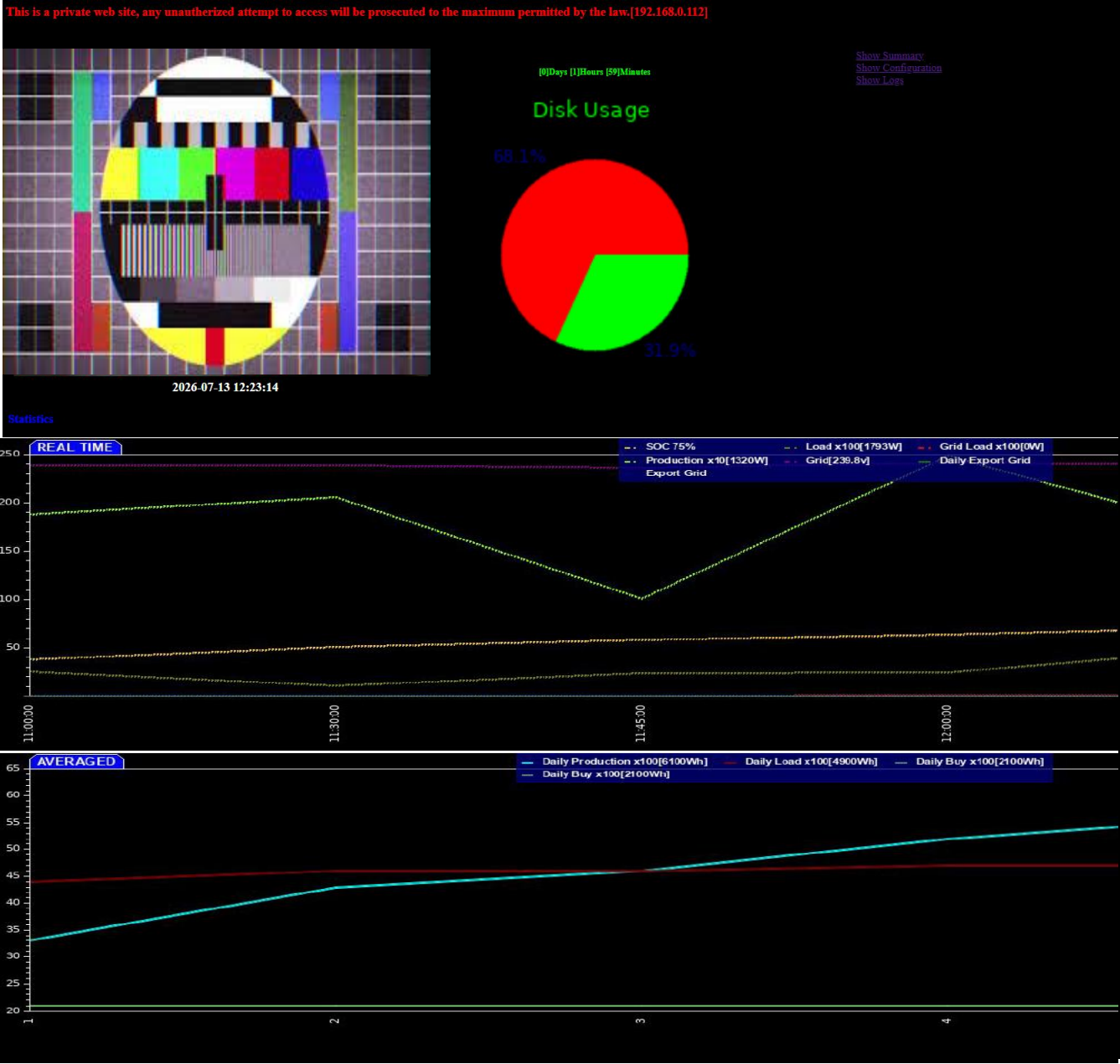
Add

ID

Remove



REBOOT



.bashrc

```
cmd=$(pgrep keep_alive.sh)
if [ -z "$cmd" ]; then
    if [ -f /home/ecsys/enable ]; then
        /home/ecsys/ecsysapp/keep_alive.sh &
    fi
fi
```

cmdline.txt

```
console=serial0,115200 console=tty1 root=PARTUUID=b64a6f0e-02
rootfstype=ext4 fsck.repair=yes rootwait ds=nocloud;i=rpi-imager-
1778722754770 cfg80211.ieee80211_regdom=IN logo.nologo
fbcon=rotate:3 vc4.force_hotplug=1 usbcore.autosuspend=-1
```

- `sudo nmcli con add con-name ethernet ifname eth0 type ethernet ip4 10.10.10.1/24 gw4 10.10.10.1`
- `sudo nmcli con mod ethernet ipv4.dns 10.10.10.1,8.8.8.8`
- `sudo nmcli con mod ethernet connection.autoconnect true`
- `sudo nmcli con up ethernet ifname eth0`

- `sudo nmcli connection add type 802-11-wireless con-name ssid_name ssid ssid_name 802-11-wireless-security.key-mgmt WPA-PSK 802-11-wireless-security.psk password`
- `sudo nmcli con mod ssid_name connection.autoconnect true`
- `sudo nmcli connection up ssid_name`

- `sudo nmcli conn delete ecsysap`
- `sudo nmcli connection add type wifi ifname wlan0 con-name "ecsysap" autoconnect yes ssid "ecsysap"`
- `sudo nmcli connection modify "ecsysap" 802-11-wireless.mode ap`
- `sudo nmcli connection modify "ecsysap" wifi-sec.key-mgmt wpa-psk`
- `sudo nmcli connection modify "ecsysap" wifi-sec.psk "ecsysabc"`
- `sudo nmcli connection modify "ecsysap" ipv4.method shared`
- `sudo nmcli connection modify "ecsysap" ipv4.addresses 10.10.10.1/24`
- `sudo nmcli connection up "ecsysap"`

`sudo apt update`

`sudo apt install dnsmasq-base`

`sudo apt remove dnsmasq`

`vi /etc/NetworkManager/conf.d/dhcp-dns.conf`

`[main]`

`dhcp=dnsmasq`

`/etc/NetworkManager/dnsmasq-shared.d/ecsysap.conf`

`dhcp-host=58:BF:25:DA:FE:73,10.10.10.2`

`dhcp-host=F8:B3:B7:8B:1C:8D,10.10.10.3`

`sudo systemctl restart NetworkManager`

`/var/www/html`

- `sudo chown www-data:www-data -R *`
- `sudo setfacl -m u:ecsys:rwX storage`

```
sudo apt-get install mariadb-server php cmake libmysqlcppconn-dev syslog-ng acl vim universal-ctags apache2 libboost-all-dev libx265-dev
sudo apt install php8.4-mysql php8.4-gd libfmt-dev ttf-mscorefonts-installer libdrm-dev libpipewire-0.3-dev libfftw3-dev
https://github.com/stephanecharette/MoveDetect
```

```
wget https://ffmpeg.org/releases/ffmpeg-8.1.2.tar.xz
wget https://github.com/opencv/opencv/archive/refs/tags/5.0.0.tar.gz
wget https://github.com/opencv/opencv_contrib/archive/refs/heads/5.x.zip
```

```
create table syscon(id MEDIUMINT,name CHAR(255) NOT NULL);
create table mask(id int NOT NULL AUTO_INCREMENT,x MEDIUMINT NULL,y MEDIUMINT NULL,w MEDIUMINT NULL,h MEDIUMINT NULL,PRIMARY KEY (id));
```

```
create table cfg(mouse_levela TINYINT NULL,mouse_levelb TINYINT,mouse_bhrs CHAR(64), mouse_name CHAR(255) NOT NULL,mouse_index MEDIUMINT
NULL,beacon_timeout MEDIUMINT NULL,dir_max MEDIUMINT NULL,night CHAR(64) NULL,ac CHAR(64) NULL,akey CHAR(64) NULL,aip CHAR(32) NULL,voice
CHAR(64) NULL,motion CHAR(64) NULL,sip CHAR(32) NULL,bip CHAR(32),sn CHAR(32) NOT NULL,bkey CHAR(32),access CHAR(255) NOT NULL, reboot TINYINT
NULL);
```

```
create table hour(noise smallint NULL,dprod smallint NULL,dload smallint NULL,dbuy smallint NULL,soc smallint NULL,upload smallint NULL,gload smallint
NULL,prod smallint NULL,gvolt smallint NULL,gdexp smallint NULL,gexp smallint NULL,ts TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP);
```

```
create user "userecsys"@"localhost"identified by "ecsys123";
grant all on ecsys.* TO "userecsys"@"localhost";
```

```
sudo nmcli dev wifi rescan
nmcli dev wifi list
sudo nmcli device wifi rescan && nmcli device wifi list
```

```
sudo systemctl --no-pager status syslog-ng
sudo systemctl restart syslog-ng
sudo systemctl enable syslog-ng
```

```
MariaDB [ecsys]> desc cfg;
```

Field	Type	Null	Key	Default	Extra
mouse_levela	tinyint(4)	YES		NULL	
mouse_levelb	tinyint(4)	YES		NULL	
mouse_bhrrs	char(64)	YES		NULL	
mouse_name	char(255)	NO		NULL	
mouse_index	mediumint(9)	YES		NULL	
beacon_timeout	mediumint(9)	YES		NULL	
dir_max	mediumint(9)	YES		NULL	
night	char(64)	YES		NULL	
ac	char(64)	YES		NULL	
akey	char(64)	YES		NULL	
aip	char(32)	YES		NULL	
voice	char(64)	YES		NULL	
motion	char(64)	YES		NULL	
sip	char(32)	YES		NULL	
bip	char(32)	YES		NULL	
sn	char(32)	NO		NULL	
bkey	char(32)	YES		NULL	
access	char(255)	NO		NULL	
reboot	tinyint(4)	YES		NULL	

```
19 rows in set (0.001 sec)
```

```
MariaDB [ecsys]> desc syscon;
```

Field	Type	Null	Key	Default	Extra
id	mediumint(9)	YES		NULL	
name	char(255)	NO		NULL	

```
2 rows in set (0.003 sec)
```

```
MariaDB [ecsys]> desc mask;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
x	mediumint(9)	YES		NULL	
y	mediumint(9)	YES		NULL	
w	mediumint(9)	YES		NULL	
h	mediumint(9)	YES		NULL	

```
5 rows in set (0.001 sec)
```

```
MariaDB [ecsys]> desc hour;
```

Field	Type	Null	Key	Default	Extra
noise	smallint(6)	YES		NULL	
dprod	smallint(6)	YES		NULL	
dload	smallint(6)	YES		NULL	
dbuy	smallint(6)	YES		NULL	
soc	smallint(6)	YES		NULL	
upload	smallint(6)	YES		NULL	
gload	smallint(6)	YES		NULL	
prod	smallint(6)	YES		NULL	
gvolt	smallint(6)	YES		NULL	
gdexp	smallint(6)	YES		NULL	
gexp	smallint(6)	YES		NULL	
ts	timestamp	NO		current_timestamp()	on update current_timestamp()

```
12 rows in set (0.001 sec)
```

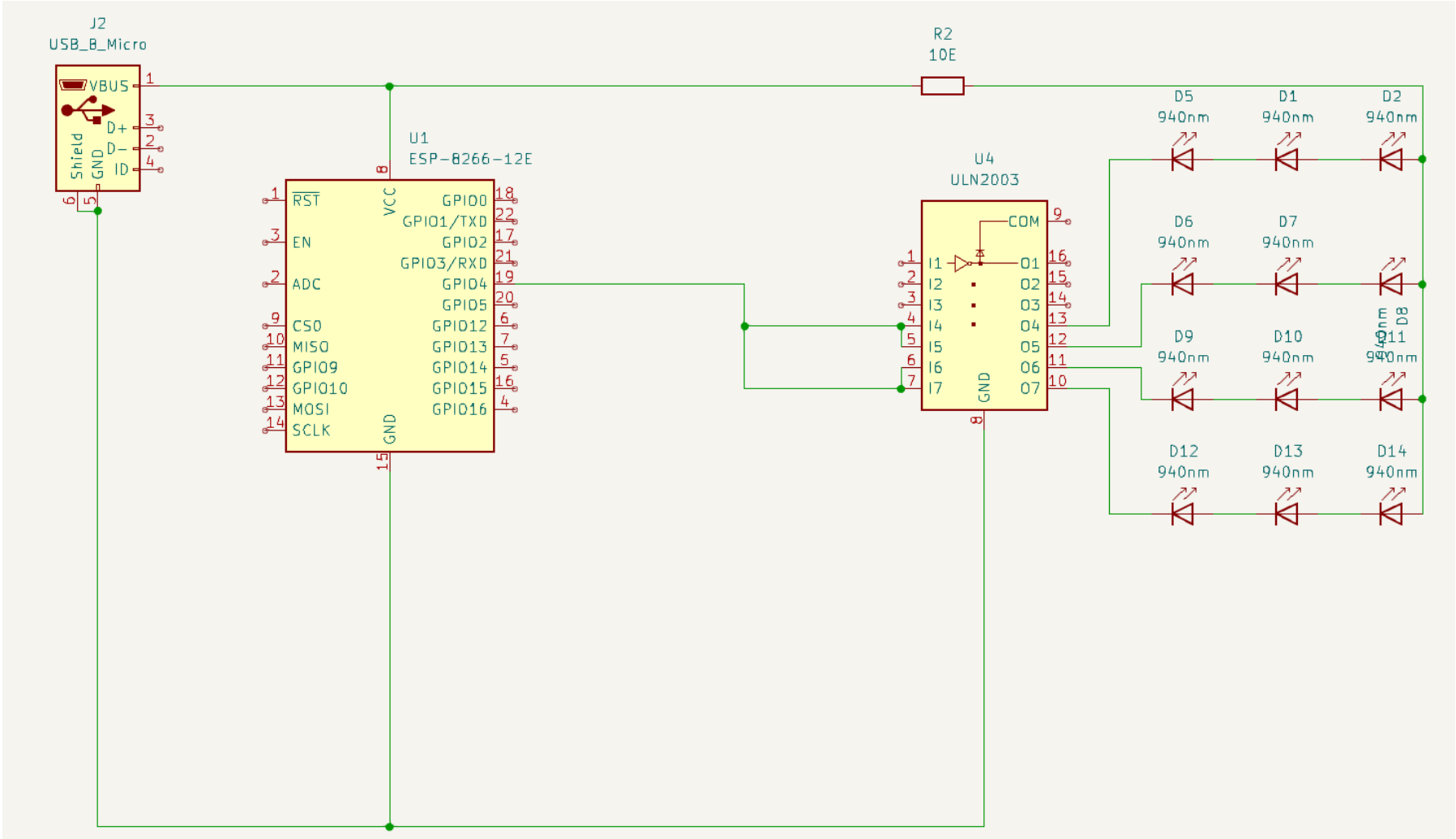
```
Database changed
```

```
MariaDB [ecsys]> show tables;
```

Tables_in_ecsys
cfg
hour
mask
out_img
syscon

```
5 rows in set (0.002 sec)
```

IR Blaster



Alarm Unit

